Multiplicative order of 2 for safe primes

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Abstract. Let $p \geq 3$ be a Sophie Germain prime, so $q = 2p+1$ is also prime. Then the order of 2 modulo q is $2p$ if $p \equiv 1 \pmod 4$ and p if $p \equiv 3 \pmod 4$. In the latter case -2 is a primitive root. This is a standard consequence of the supplementary law of quadratic reciprocity, not evidence that infinitely many Sophie Germain primes exist.

Sequence and scope. OEIS A005384 lists primes p for which $q = 2p + 1$ is also prime [1]. The result treats every such odd p, including $p = 3$; the exceptional $p = 2$ has $q = 5$ and $\mathrm{ord}_5(2) = 4$. A published treatment of primitive roots at safe primes appears in [2].

Theorem 1. *Suppose $p \geq 3$ and both p and $q = 2p + 1$ are prime. Then*

$$\mathrm{ord}_q(2) = \begin{cases} 2p, & p \equiv 1 \pmod 4, \\ p, & p \equiv 3 \pmod 4. \end{cases}$$

Furthermore 2 is a primitive root modulo q in the first case and -2 is a primitive root modulo q in the second.

Proof. Fermat's theorem gives $2^{2p} \equiv 1 \pmod q$. Since $q \geq 7$, neither 2 nor $2^2 = 4$ is 1 modulo q. The order of 2 thus divides $2p$ but is neither 1 nor 2. Because p is prime, its only remaining possibilities are p and $2p$.

By Euler's criterion and the supplementary law $(2/q) = (-1)^{(q^2-1)/8}$ [2],

$$2^p = 2^{(q-1)/2} \equiv \begin{cases} -1 \pmod q, & q \equiv 3 \pmod 8, \\ 1 \pmod q, & q \equiv 7 \pmod 8. \end{cases}$$

As $q = 2p + 1$, these two congruence classes correspond respectively to $p \equiv 1$ and $p \equiv 3 \pmod 4$, proving the formula. In the latter case, p is odd and $2^p \equiv 1 \pmod q$, hence $(-2)^p \equiv -1 \pmod q$; the order of -2 divides $2p$ but cannot be 1, 2, or p, because $q \geq 7$ and $(-2)^2 = 4 \not\equiv 1 \pmod q$. It must be $2p$, so -2 is primitive. □

Checks. At $p = 5$, $q = 11$ and $\mathrm{ord}_{11}(2) = 10$; at $p = 11$, $q = 23$ and $\mathrm{ord}_{23}(2) = 11$ while -2 has order 22.

Priority and verification note. This is an established, routine safe-prime fact, not a proof of any open infinitude conjecture associated with A005384. Calculations and proof steps were AI-checked; independent human review and novelty are not claimed.

References. [1] N. J. A. Sloane et al., *The On-Line Encyclopedia of Integer Sequences*, A005384, `https://oeis.org/A005384` (accessed 12 Nov. 2026). [2] D. Pierce, *Elementary Number Theory*, Theorem 90 (primitive roots for Germain primes), Mathematics Department lecture notes, 2012, `https://mat.msgsu.edu.tr/~dpierce/Dersler/Number-theory/2012/number-theory-text-2012.pdf`.